Exploring variation in the exponential metalaw of physics

Work in progress: this writeup is still being updated.

Abstract

The choice of operators and functional forms in scientific modelling is often informed by the domain. Quantifying these preferences might help construct priors that guide symbolic regression. Previous work found that ranked operator frequencies in physics equations approximately follow exponential laws, with a lower fitted decay rate for Encyclopaedia Inflationaris than for three broader physics collections. We revisit this difference as a starting point for exploring whether operator statistics vary across physical domains. Using log least squares and whole-equation bootstrap resampling, we compare equation collections and seven branches within an expanded Feynman corpus. We test proposed explanations for the earlier difference by changing representation conventions, constructing a comparison collection with similar scope, and examining variation across physics branches. The results show variation in fitted rates across the selected collections and branches, but do not isolate a single physical explanation; some estimates are sensitive to how equations are classified. The analysis is limited by its selection of algebraic expressions, the exponential model's imperfect description of rare-operator tails, and approximate uncertainty estimates. We discuss how these observations, together with per-expression patterns of decay rate and complexity, could inform domain-dependent penalties for symbolic regression. Their ability to improve equation recovery or search efficiency remains to be evaluated.

1 Introduction

When fitting equations to scientific data, the choice of candidate operators is often informed by the domain. Dilogarithms, for example, may be a natural choice in particle physics, whereas they would be a less usual starting point for modelling biological systems. Expectations also extend to how operators are combined. Expressions in special relativity often involve the Lorentz factor, while there is no comparable general expectation for thermodynamic equations. Such differences suggest that scientific domains may exhibit characteristic patterns in both operator usage and the structure of expression trees.

Quantifying these patterns could help inform priors for symbolic regression. A prior adapted to the relevant domain could favour candidate expressions with structures commonly encountered in that domain, while allowing less familiar forms. Such information might constrain the search space or guide the search more efficiently towards the Pareto front of predictive accuracy and expression complexity. This motivates the study of statistical patterns in existing equations and how those patterns vary across branches of science.

Previous work [1] has shown that the ranked frequencies of mathematical operators in collections of physics equations can be described approximately by exponential laws, in contrast to Zipf's power law for word frequencies in natural language. Each equation is represented as an expression tree, and occurrences of each operator type are counted across the collection. The types are then ranked from most to least frequent. The counting convention includes variables, constants and numerical literals alongside arithmetic operations and named functions.

The normalised exponential model takes the form

$$p_r(\beta) = \frac{e^{-\beta r}}{\sum_{s=1}^{R} e^{-\beta s}}, \qquad r = 1, \ldots, R, \tag{1}$$

where R is the number of observed operator types. The decay rate β describes how rapidly frequency decreases with rank. A larger value corresponds to a distribution more concentrated among the most frequent types.

The original study found similar fitted decay rates for the Feynman benchmark, Wikipedia equations and the Cambridge Handbook of Physics Formulas, but a lower rate for Encyclopaedia Inflationaris [1]. Understanding

this difference is relevant to the construction of priors for symbolic regression. A common statistical description across physics could support a shared component of such a prior, whereas systematic variation with physical domain could motivate conditioning it on the branch under study.

We therefore begin by revisiting the inflationary difference before comparing individual branches of physics. The proposed explanations for the gap concern the treatment of physical constants, the inflationary collection's focus on one physical concept, the hypothetical nature of its models, and the energy scales involved [1]. We estimate decay rates using log least squares and obtain individual uncertainty intervals by resampling whole equations. Applying the same procedure across the collections allows us to examine how the estimates change under the comparisons.

First, we investigate constant conventions by setting $c = \hbar = 1$ in textbook expressions and removing the common amplitude prefactor M^4 from the inflationary potentials. These changes alter the expression trees and their operator counts. We examine whether they account for the difference in fitted rates.

Second, we compare the inflationary collection with proposed models from modified gravity. Encyclopaedia Inflationaris assembles forms of the inflaton potential in single-field slow-roll inflation [2]. It therefore concerns one physical quantity and contains hypothetical models proposed to describe inflation. Our comparison collection contains proposed curvature functions from $f(R)$ gravity, in which the gravitational action depends on a function of the scalar curvature R [3]. This collection shares both features, allowing us to ask whether another collection of hypothetical models centred on one physical concept has a similar fitted rate. The comparison does not separate the effects of those two features. Section 2 and Appendix A identify the source papers and expressions.

Third, the proposed energy-scale explanation motivates a broader comparison across physics branches. We expand the 100-expression AI Feynman benchmark [4] using expressions from all three volumes of the publicly accessible Feynman Lectures [5]. We compare seven branches and examine alternative assignments for chapters spanning more than one branch. These comparisons explore variation with physical domain, although branch labels do not isolate energy scale from other differences between the equations.

This project makes three contributions. First, it reassesses the inflationary difference using a consistent fitting procedure and individual bootstrap uncertainty estimates. Second, it assembles an expanded Feynman collection with source references and branch labels, together with a catalogue of 37 modified-gravity expressions. Third, it examines variation in fitted rates under changes to constant conventions, collection scope and branch assignment. These analyses provide an exploratory basis for considering domain-dependent priors; their effect on symbolic-regression performance remains to be evaluated.

Section 2 describes the collections, operator counting, fitting procedure and uncertainty estimates. Section 3 revisits the inflationary difference through the constant-convention and modified-gravity comparisons. Section 4 examines variation across physics branches. Section 5 discusses the findings, their limitations and the prospect of using expression statistics to guide symbolic regression. The appendices document catalogue sources, sensitivity checks, bootstrap calibration and the exponential limit of the Zipf–Mandelbrot distribution.

2 Data and methods

2.1 Equation collections

We begin with three collections released with the original study [1]: 100 expressions from the AI Feynman benchmark, 41 equations from Wikipedia's list of scientific equations named after people [6], and 70 inflaton potentials from Encyclopaedia Inflationaris [2]. We extend the Wikipedia collection with 35 algebraic expressions, bringing its total to 76. As a mathematical comparison, we also include 61 probability density or mass functions from Leemis and McQueston [7].

For the comparison across physics branches, we expand the Feynman benchmark using expressions from all three volumes of the Feynman Lectures [5]. After selection and deduplication, this adds 916 expressions. The combined collection contains 1016 expressions, of which 968 are assigned to the seven branches considered here. Because the original benchmark is part of the expanded collection, the two collections overlap.

The modified-gravity collection contains 37 proposed curvature functions from the review and model papers listed in Appendix A. These expressions include related variants: 17 come from the model families examined by Pérez-Romero and Nesseris [8]. The collection therefore contains 37 expressions, which should not be interpreted as 37 independent model families. Each catalogue entry records its expression and source, and Appendix A provides the corresponding source table.

2.2 Equation selection and operator counting

For the additional Feynman Lectures expressions, we select equations that give one quantity as an explicit algebraic expression. We count operators on the right-hand side. We exclude expressions consisting of a single symbol or number, as well as those containing derivatives, integrals, sums, limits, operator hats or complex conjugates.

Each expression is represented as a tree, following the counting conventions of the original study [1]. Arithmetic operations and named functions contribute operator occurrences. Variables, physical constants and numerical literals are also counted, each as a separate category of tree leaves. Powers x^2, x^3 and x^4 are represented by distinct unary operators. Appendix A gives further transcription and counting conventions.

We assign expressions from the Lectures to physics branches according to their source chapters. Benchmark expressions retain their existing labels, apart from the relativistic-electrodynamics adjustment described in Appendix A. Combining fluid and continuum mechanics gives seven branches: classical mechanics, special relativity, electromagnetism, waves and optics, thermodynamics and kinetic theory, quantum mechanics, and fluid and continuum mechanics.

The other 48 expressions remain in the expanded collection but are excluded from the seven-branch comparison. They comprise 31 mathematics expressions, nine probability expressions, six general-relativity expressions and two without a branch assignment. Some chapters span more than one branch; we examine alternative assignments for these chapters as a sensitivity check.

2.3 Fitting the exponential decay rate

For each collection or branch, we count occurrences of every operator type and arrange the positive counts in descending order,

$$n_1 \geq n_2 \geq \cdots \geq n_R > 0.$$

The observed frequency at rank r is

$$f_r = \frac{n_r}{N}, \qquad N = \sum_{r=1}^{R} n_r.$$

We fit the normalised exponential model introduced in Eq. (1) by minimising the squared differences between observed and model log frequencies:

$$Q(\beta) = \sum_{r=1}^{R} \left[\log f_r - \log p_r(\beta)\right]^2 . \tag{2}$$

Each observed rank receives equal weight in this criterion. We include every operator type with a positive count. Types absent from a collection are omitted because their log frequencies are undefined.

The ranking is constructed separately for each collection or branch. Consequently, both the number of observed types and the operator occupying a particular rank can differ between groups. The fitted rate $\hat{\beta}$ summarises how concentrated each ranked distribution is; it does not describe which operators are most common.

We constrain the numerical optimisation to $0 \leq \beta \leq 5$ and flag fits near either boundary. Calculating this fitted summary does not require operator occurrences within an equation to be independent. The assumptions needed to interpret its uncertainty enter through the resampling procedure below.

2.4 Uncertainty from equation resampling

We estimate uncertainty by resampling whole equations within each collection or branch. Each bootstrap sample contains the same number of equations as the original group, drawn with replacement. An equation can therefore appear more than once in a sample, while another may be absent.

For each sample, we recount the operators, determine which types are present, rank their frequencies and refit β. Repeating this procedure 10000 times gives a distribution of fitted rates. We report the fit to the original collection, the standard deviation of the bootstrap estimates as a standard error, and their 2.5th and 97.5th percentiles as an individual 95% bootstrap interval.

Resampling whole equations keeps together the operators that occur within each expression. Interpreting the resulting intervals as sampling uncertainty assumes that the equations are sufficiently independent and

represent a wider collection of comparable expressions. Related equations or variants of the same model can limit this assumption. The intervals also do not account for systematic omissions or transcription errors in the collections.

The 95% label is approximate. Appendix C repeats the procedure on artificial collections whose true decay rate is known. The resulting interval contains that rate less often than the label suggests. We report each interval individually and do not widen the intervals in the tables to compensate.

We also resample whole source chapters to examine dependence among equations from the same chapter. This check, together with alternative branch assignments and selection rules, is described in Appendix B.

For reproducibility, we use random seed 20261013 and calculate percentiles by linear interpolation (Hyndman–Fan type 7). Failed and degenerate fits are recorded, and intervals affected by missing fits are marked unresolved.

3 Revisiting the inflationary difference

3.1 Comparison of the fitted rates

We first examine whether the lower decay rate reported for Encyclopaedia Inflationaris is reproduced using the fitting procedure described in Section 2. Table 1 and Figure 1 summarise the estimates for all collections.

The Feynman benchmark gives a fitted rate of $\hat{\beta} = 0.342$, with an individual 95% bootstrap interval of $[0.319, 0.371]$. Encyclopaedia Inflationaris gives $\hat{\beta} = 0.269$, with an interval of $[0.253, 0.293]$. The point estimates differ by 0.073, and their individual intervals do not overlap. The lower inflationary estimate is therefore reproduced with log least squares and equation resampling.

A lower decay rate corresponds to a slower decline in frequency with rank. Under the fitted exponential model, operator occurrences are therefore less concentrated among the most frequent types in the inflationary collection than in the Feynman benchmark. This describes a difference between their ranked distributions, without identifying its physical origin.

Corpus	Expr.	Instances	Ranks	$\hat{\beta}$	Boot. SE	Individual interval
Feynman benchmark	100	1167	19	0.342	0.013	[0.319, 0.371]
Feynman benchmark + Lectures	1016	10693	22	0.354	0.008	[0.333, 0.366]
Wikipedia	76	1006	17	0.315	0.015	[0.284, 0.346]
Encyclopaedia Inflationaris	70	1376	25	0.269	0.010	[0.253, 0.293]
Leemis	61	1044	24	0.269	0.014	[0.250, 0.304]
f(R)	37	650	24	0.270	0.021	[0.264, 0.347]

Table 1: Corpus estimates from log least squares. Intervals are nominal individual 95% equation-bootstrap intervals. The instances column counts all operator occurrences, including nullary types.

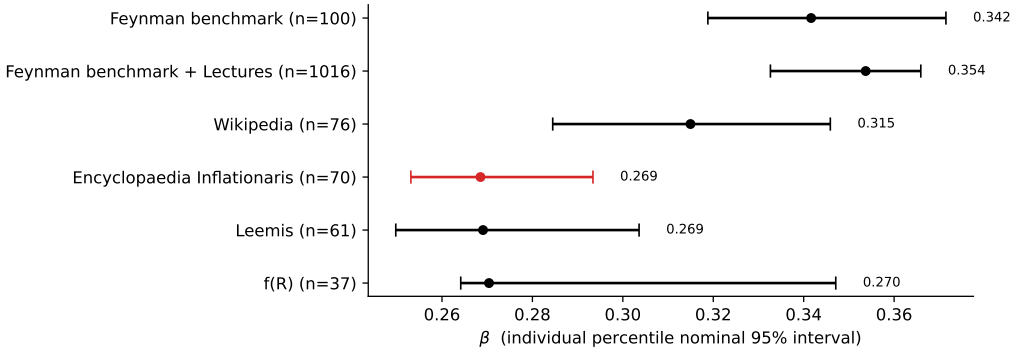


Figure 1: Corpus estimates and nominal individual 95% equation-bootstrap intervals.

3.2 Changes to constant conventions

One proposed explanation concerns the conventions used to write the equations. Physical constants and amplitude prefactors contribute to the expression trees, so removing them can change the operator counts and the fitted decay rate. We examine two such changes.

First, we set $c = \hbar = 1$ in the textbook expressions. This affects 32 expressions in the Feynman benchmark and lowers its fitted rate from 0.342 to 0.332. Only two expressions in the Wikipedia collection are affected; its rate rounds to 0.315 both before and after the change.

Second, we remove the common amplitude prefactor M^4 from the inflationary potentials by setting $M = 1$. This affects all 70 potentials and lowers the fitted rate from 0.269 to 0.261.

After both changes, the difference between the Feynman benchmark and the inflationary collection is 0.071, compared with 0.073 before the changes. These conventions therefore account for little of the observed difference in point estimates. Other choices of algebraic representation may also affect operator counts, but they are not examined in this comparison.

3.3 Comparison with modified gravity

We next consider the proposed explanations involving the inflationary collection's focus on one physical concept and its inclusion of hypothetical models. The $f(R)$ gravity collection shares these features: its expressions are proposed forms of a curvature function entering the gravitational action.

The 37 modified-gravity expressions give a fitted rate of $\hat{\beta} = 0.270$, with an individual 95% bootstrap interval of $[0.264, 0.347]$. Its point estimate is close to the inflationary value of 0.269, and the intervals overlap. A similarly low fitted rate is therefore found in another collection of hypothetical models centred on one physical concept.

This similarity does not establish why the rates are close. Both features change together in the comparison, so their effects cannot be distinguished. The modified-gravity collection also contains related variants of model families. These variants influence the operator frequencies and limit the independence assumption underlying equation resampling.

Together, the comparisons reproduce the lower inflationary estimate and show that it changes little relative to the Feynman benchmark under the tested constant conventions. The modified-gravity collection provides a further example of a similar fitted rate, while leaving its explanation unresolved. We next examine variation across physics branches within the expanded Feynman collection.

4 Variation across physics branches

The original study proposed differences in energy scale as one possible explanation for the lower inflationary decay rate [1]. Inflation also concerns a different branch of physics from much of the material in the broader equation collections. This motivates us to examine whether fitted decay rates vary across physics branches. We use the expanded Feynman collection because the expressions in the Feynman Lectures are publicly accessible [5]. Branch labels provide a way to explore variation with physical domain, although they do not isolate the effect of energy scale.

The fitted rates show a pattern suggestive of two broad groups (Table 2 and Figure 2). Quantum mechanics, electromagnetism, waves and optics, and classical mechanics have estimates between 0.325 and 0.360, with overlapping individual bootstrap intervals. Thermodynamics and kinetic theory and fluid and continuum mechanics have higher estimates, at 0.403 and 0.422, and their intervals also overlap.

The separation between these groups is incomplete. Special relativity has a relatively wide interval that overlaps estimates in both groups. The plot therefore suggests a grouping of the fitted rates, without establishing a sharp division between branches. Figure 2 also includes the original Feynman benchmark, the full expanded Feynman collection, and the modified-gravity, inflationary and Wikipedia collections for comparison.

Branch	Expr.	Instances	Ranks	$\hat{\beta}$	Boot. SE	Individual interval
Special relativity	89	1073	14	0.382	0.028	[0.322, 0.432]
Quantum mechanics	221	2374	17	0.327	0.016	[0.286, 0.349]
Electromagnetism	312	3285	18	0.325	0.010	[0.313, 0.353]
Waves and optics	126	1448	15	0.346	0.017	[0.303, 0.367]
Classical mechanics	105	1014	15	0.360	0.020	[0.297, 0.377]
Thermodynamics and kinetic theory	73	727	16	0.403	0.021	[0.371, 0.452]
Fluid and continuum mechanics	42	360	13	0.422	0.027	[0.377, 0.483]

Table 2: Branch estimates and nominal individual 95% equation-bootstrap intervals.

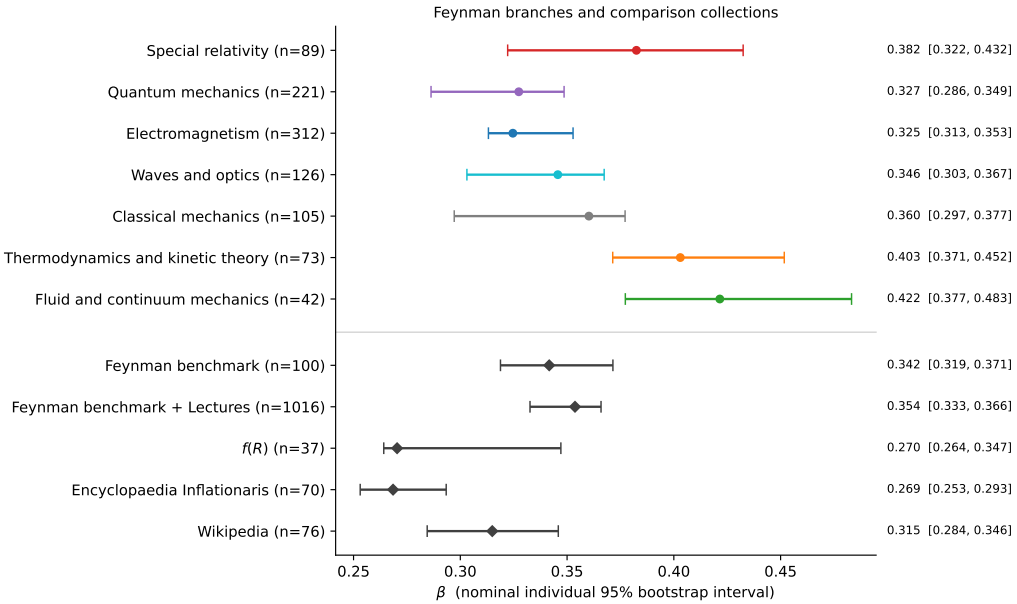


Figure 2: Fitted rates for the seven Feynman branches (circles) and five comparison collections (diamonds, below the dividing line), with nominal individual 95% equation-bootstrap intervals. The comparison collections are the 100-expression Feynman benchmark, the full expanded Feynman collection, $f(R)$ gravity, Encyclopaedia Inflationaris and Wikipedia. The benchmark is contained in the expanded collection, whose expressions also supply the branch estimates.

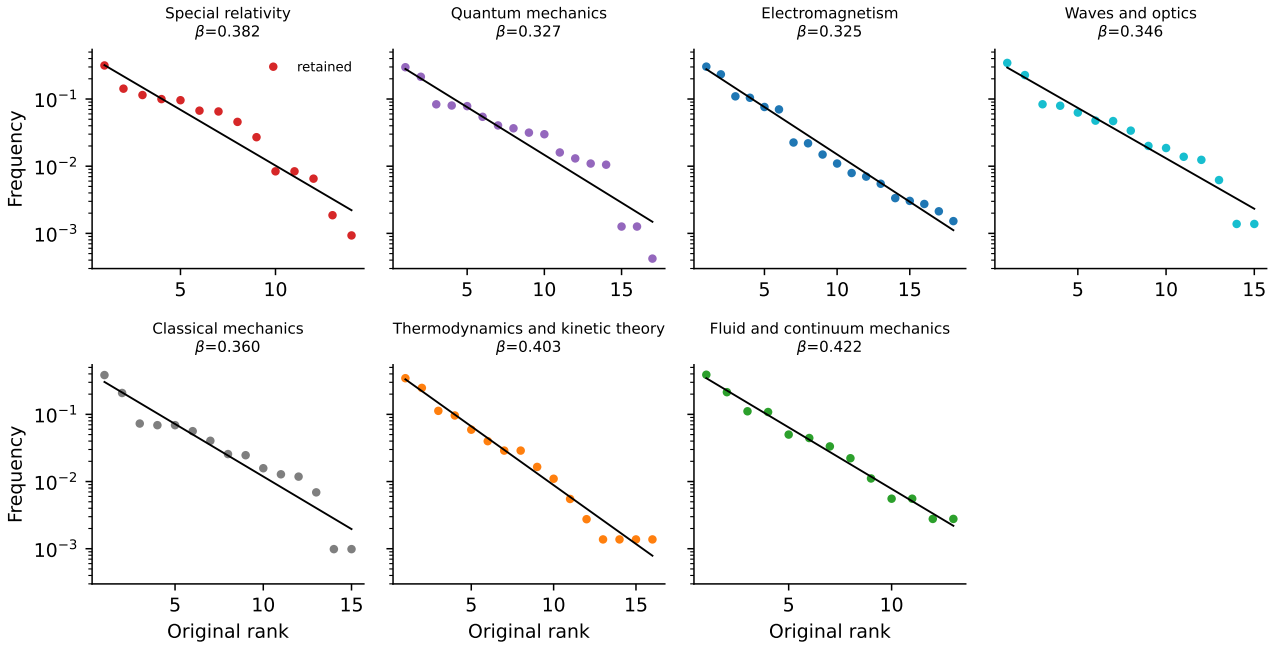


Figure 3: Observed rank–frequency distributions and fitted exponentials for the seven branches.

Figure 3 shows the distributions underlying these estimates. Each branch is ranked separately, and both the observed operator types and their ordering can differ. The fitted rates describe how concentrated the ranked frequencies are; identifying the operators responsible for these patterns requires examining their identities separately.

Additional checks examine the effects of branch classification and relationships between equations from the same chapter. The special-relativity estimate changes substantially when ambiguously classified chapters are removed or reassigned. Resampling whole chapters widens the quantum- and fluid-mechanics intervals, while leaving them separated. Appendix B gives the procedures and results. These checks help distinguish patterns in the selected expressions from conclusions that would require broader evidence about each physics branch.

5 Discussion

5.1 Interpretation of the comparisons

This study reproduces the lower fitted decay rate of Encyclopaedia Inflationaris relative to the Feynman benchmark. Setting $c = \hbar = 1$ in the textbook expressions and removing the inflationary amplitude prefactor leave most of the difference intact. These particular conventions therefore account for little of the observed gap.

The modified-gravity collection gives a rate close to the inflationary value. Both collections contain proposed models centred on one physical concept. Their similar rates are consistent with explanations involving these features, although the comparison cannot distinguish their effects or establish either as the cause.

Within the expanded Feynman collection, the estimates suggest a broad grouping. Quantum mechanics, electromagnetism, waves and optics, and classical mechanics have lower fitted rates than thermodynamics and fluid mechanics. This pattern motivates further investigation of how mathematical structure varies across physical domains. It does not directly establish an energy-scale explanation, because the branches also differ in their subject matter, quantities of interest and equation forms.

5.2 Limitations of the analysis

A major limitation is the restricted form of the expressions considered. In particular, the additional Feynman collection includes equations that give one quantity as an explicit algebraic expression. Differential and integral forms are excluded, together with expressions containing sums, limits, operator hats or complex conjugates.

These restrictions omit substantial parts of the mathematical language used in physics. The resulting operator statistics therefore describe the selected algebraic expressions rather than the full range of equations used in each branch.

The exponential model also does not capture the flat tail of the observed distributions well. When several operator types occur only once or twice, their ranked frequencies form plateaus, whereas an exponential with positive decay rate decreases continuously. Because log least squares gives each rank equal weight, these rare types can influence the fitted rate despite accounting for few operator occurrences. Adding or removing equations can change which rare operators appear and how often they occur, thereby changing both the tail and the fitted value. The estimated rate consequently depends on the composition of the collection as well as the adequacy of the exponential approximation.

Other choices, including algebraic representation, branch assignment and the number of variants retained from a model family, also affect the counts. The substantial change in the special-relativity estimate under alternative classifications illustrates this sensitivity.

The bootstrap intervals describe variation under resampling of the selected expressions. They do not capture uncertainty arising from omitted equation formats, systematic selection choices or transcription errors. Related equations and model variants can also limit the assumption that equations provide independent sampling units. Chapter resampling examines some of this dependence, although certain branches contain few chapters.

We also checked whether an interval labelled 95% deserves that label. Appendix C builds artificial collections in which every operator frequency follows an exponential with true rate 0.35. For each collection we calculate the same interval reported in the tables and record whether 0.35 lies inside it. In 200 collections, a genuine 95% interval would contain the true rate about 190 times. In the simplest collections it did so 170 to 187 times. When equations from the same chapter were tied together and whole chapters were resampled, it did so only 149 to 169 times. The interval is therefore often too short, and two intervals that fail to overlap are weaker evidence of a difference than the label suggests. The check uses invented operator counts rather than real equations, and the intervals reported above have not been widened to match it.

5.3 Per-expression patterns and symbolic regression

The expression catalogue provides a complementary view by fitting a decay rate to each expression individually and plotting it against expression complexity. For an expression g, the fitted value $\hat{\beta}_g$ is calculated from its own ranked operator frequencies, while its complexity k_g is the number of nodes in its expression tree. These values describe individual expressions, whereas the corpus rate is obtained by pooling operator counts across the collection.

The apparent concentration of expressions in particular regions of the β–complexity plot suggests a possible way to guide symbolic regression. Rather than requiring a candidate expression to resemble the pooled distribution of an entire corpus, we could assess whether its combination of fitted rate and complexity is typical of expressions in the relevant domain.

One possible objective would take the form

$$\mathcal{L}(g) = \mathcal{L}_{\text{data}}(g) + \lambda\, \mathcal{R}_d(\hat{\beta}_g, k_g), \tag{3}$$

where $\mathcal{R}_d$ penalises combinations of fitted rate and complexity that are unusual in domain d. Conditioning on complexity would allow the preferred range of β to vary with expression size. This could provide an initial structural preference that can be evaluated directly for each candidate expression.

Before constructing such a penalty, the origin of the apparent regions needs to be examined. Short expressions contain few operator occurrences, and their integer counts permit only a limited set of frequency patterns. Some bands or concentrations may therefore arise from expression size and tree structure alone. The catalogue's comparison with random expression trees of the same complexities provides a starting point for investigating this possibility. Further comparisons could examine whether the observed patterns persist after accounting for complexity, operator vocabulary and the number of distinct types.

The usefulness of this information remains to be tested. A penalty learned from one set of expressions could be evaluated on separate symbolic-regression problems with known generating equations, comparing recovery rates and computational effort with an unmodified search. This would establish whether the per-expression patterns provide useful guidance beyond a conventional preference for simpler expressions.

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A Catalogue sources and transcription conventions

The catalogue records the expression and its source reference. Lectures entries include volume, chapter and equation identifiers. The additional Wikipedia entries follow the original release cutoff in the same alphabetical list. The source references and counting metadata are retained with the individual records.

Physical constants such as c, $\hbar$ and G are classified as constants. The symbol q_e retains its classification as a variable in the benchmark. Time averages written as $\langle x \rangle$ are represented by a single symbol. In the modified-gravity catalogue, R denotes curvature rather than the gas constant, and model parameters are counted as variables. The catalogue stores the full curvature function: when a source writes $F(R) = R + f(R)$, the correction $f(R)$ is combined with R. Optional additional terms proposed later in a source are not included in the defining expression unless that expression itself contains them.

For branch assignment, the benchmark expression from the relativistic-electrodynamics chapter in Volume I is assigned to special relativity. The corresponding Lectures chapters—Volume I, Chapter 34, and Volume II, Chapters 25, 26 and 28—receive the same classification. The catalogue records ambiguous assignments for the sensitivity analysis.

Table 3 identifies the source used to transcribe each modified-gravity expression. Where a review is cited, it is the transcription source; the citation does not attribute the original proposal of every model to that review. The entry identifiers match those in the catalogue.

Table 3: Transcription sources for the 37 modified-gravity expressions.

Catalogue entry	Source
`fr.power`	[3]
`fr.inverse_power`	[3]
`fr.quadratic`	[3]
`fr.inverse_plus_quadratic`	[3]
`fr.chameleon`	[3]
`fr.logarithm`	[9]
`fr.exponential`	[10]
`fr.sine`	[3]
`fr.two_scales`	[3]
`fr.starobinsky_2007`	[3]
`fr.hu_sawicki`	[11]
`fr.tsujikawa`	[12]
`fr.exponential_gravity`	[13]
`fr.appleby_battye`	[14]
`fr.miranda_log`	[15]
`fr.rational_nojiri`	[16]
`fr.two_tanh`	[16]
`fr.odd_rational`	[16]
`fr.kruglov_arctan`	[17]
`fr.kruglov_arctan_quadratic`	[18]
`fr.pn_sqrt`	[8]
`fr.pn_inverse`	[8]
`fr.pn_three_halves`	[8]
`fr.pn_arctanh`	[8]
`fr.pn_sin`	[8]
`fr.pn_sinh`	[8]
`fr.pn_exp`	[8]
`fr.pn_tanh`	[8]
`fr.pn_asin`	[8]
`fr.pn_log`	[8]
`fr.pn_tan`	[8]
`fr.pn_arctan`	[8]

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Table 3 (continued)

Catalogue entry	Source
`fr.pn_erf`	[8]
`fr.pn_asinh`	[8]
`fr.pn_polynomial`	[8]
`fr.pn_pade`	[8]
`fr.pn_hypergeometric`	[8]

B Sensitivity checks

B.1 Source-chapter resampling

We resample whole chapters jointly across branches. Each chapter is assigned to a stratum corresponding to the branch containing most of its retained equations, with ties resolved using the recorded branch order. Within each stratum, we sample the original number of chapters with replacement. Every equation from a selected chapter receives the same multiplicity, including equations assigned to other branches. This preserves dependence induced by the 11 chapters represented in more than one branch; the number of equations per branch can vary between replicates.

The individual chapter-bootstrap interval is [0.375, 0.523] for fluid and continuum mechanics and [0.279, 0.356] for quantum mechanics. Both are wider than their equation-bootstrap intervals. The four chapters represented in fluid and continuum mechanics provide a small number of resampling units, so this check gives limited information about its uncertainty.

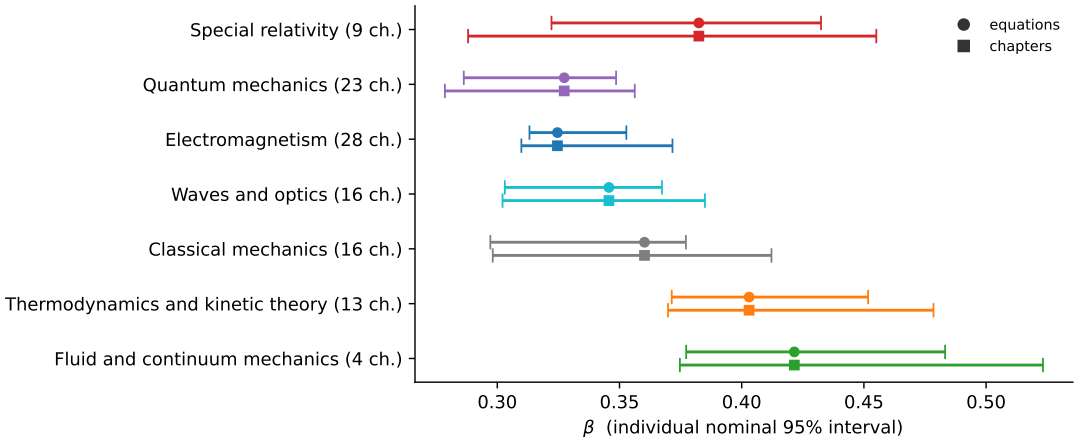


Figure 4: Equation and joint chapter bootstrap intervals at the nominal individual 95% level. Labels give the number of source chapters represented in each branch.

B.2 Branch assignments and selection

The alternative specifications remove chapters with ambiguous assignments, reassign those chapters, remove expressions flagged for averages or imaginary units, or remove Volume III. They use the same log-least-squares fitting procedure. Table 4 reports the estimates and individual equation-bootstrap intervals; the original branch assignment remains the main analysis.

Specification	Branch	Expr.	$\hat{\beta}$	Individual interval
Ambiguous chapters removed	SR	32	0.231	[0.204, 0.289]
	QM	151	0.320	[0.284, 0.350]
	EM	198	0.315	[0.302, 0.351]
	Waves	48	0.291	[0.265, 0.341]
	Classical	75	0.354	[0.298, 0.390]
	Thermo	61	0.405	[0.368, 0.461]
	Fluid	42	0.422	[0.377, 0.483]
Ambiguous chapters reassigned	SR	32	0.231	[0.204, 0.289]
	QM	182	0.335	[0.299, 0.364]
	EM	343	0.336	[0.315, 0.359]
	Waves	63	0.303	[0.283, 0.345]
	Classical	75	0.354	[0.298, 0.388]
	Thermo	89	0.353	[0.321, 0.388]
	Fluid	44	0.419	[0.374, 0.478]
Notation-flagged expressions removed	SR	88	0.395	[0.314, 0.437]
	QM	164	0.317	[0.284, 0.341]
	EM	273	0.330	[0.318, 0.361]
	Waves	99	0.362	[0.320, 0.393]
	Classical	97	0.365	[0.309, 0.386]
	Thermo	69	0.400	[0.368, 0.449]
	Fluid	39	0.422	[0.380, 0.500]
Volume III removed	SR	89	0.382	[0.321, 0.431]
	QM	26	0.349	[0.319, 0.426]
	EM	312	0.325	[0.313, 0.353]
	Waves	126	0.346	[0.303, 0.368]
	Classical	105	0.360	[0.296, 0.377]
	Thermo	73	0.403	[0.370, 0.451]
	Fluid	42	0.422	[0.378, 0.481]

Table 4: Alternative branch assignments and selection rules. Each entry reports the fitted rate and its nominal individual 95% equation-bootstrap interval. Waves denotes waves and optics; fluid denotes fluid and continuum mechanics.

C Bootstrap calibration

We examine interval coverage using 200 synthetic datasets in each of three scenarios, each with seven branches and 400 bootstrap replicates. All scenarios have population marginal operator frequencies following an exponential with rate $\beta = 0.35$. The first uses identical distributions and independent equation count vectors. The second varies equation counts, operator vocabularies and expression lengths, with dependence within an equation introduced through a shared random probability vector. The third introduces a shared random probability vector within each chapter and resamples whole chapters, including a branch with only four chapters.

For each branch, we record whether the individual interval contains the known rate 0.35. Out of 200 collections, it does so between 170 and 189 times when equations are resampled, and between 149 and 169 times when whole chapters are resampled. A 95% interval would contain the true rate about 190 times, so these intervals are often too short. The shortfall is largest in the chapter-dependent scenario. We also record failed fits. The artificial counts are not expression trees. The check identifies the shortfall; it does not replace the intervals reported for the observed collections.

D Exponential limit of the Zipf–Mandelbrot distribution

For a fixed set of ranks $r = 1, \dots, R$, the normalised Zipf–Mandelbrot distribution is

$$p_r(\alpha, b) = \frac{(r+b)^{-\alpha}}{\sum_{s=1}^{R}(s+b)^{-\alpha}}, \qquad \alpha > 0, \quad b > 0. \tag{4}$$

An exponential distribution is obtained when $b \to \infty$ and $\alpha/b \to \beta > 0$. To see this, cancel the common factor $b^{-\alpha}$ from the numerator and denominator:

$$p_r(\alpha, b) = \frac{\exp[-\alpha \log(1 + r/b)]}{\sum_{s=1}^{R} \exp[-\alpha \log(1 + s/b)]}. \tag{5}$$

Since R is fixed, the expansion

$$\alpha \log(1 + r/b) = \frac{\alpha}{b} r + O\left(\frac{\alpha R^2}{b^2}\right) \tag{6}$$

holds uniformly over the ranks. Under the specified scaling, the remainder tends to zero and hence

$$p_r(\alpha, b) \longrightarrow \frac{e^{-\beta r}}{\sum_{s=1}^{R} e^{-\beta s}}, \tag{7}$$

recovering Eq. (1). The scaling of both parameters matters: if b increases while α is held fixed, the limiting distribution is uniform, $p_r \to 1/R$.

This limiting relationship helps interpret comparisons of the Zipf–Mandelbrot and exponential models. Whether fitted parameters for a particular corpus approach this regime requires examining their values relative to the observed rank range. The limiting argument alone makes no claim about the empirical preference between the models.